Practice Set 2 – Hospital Database

Q1. Create the hospital database and both tables exactly as shown.

```
121 •   USE hospital;
122
123 •   CREATE TABLE doctors (
124         doctor_id INT PRIMARY KEY,
125         dname VARCHAR(50) NOT NULL,
126         department VARCHAR(30)
127     );
128
129 •   CREATE TABLE appointments (
130         appt_id INT PRIMARY KEY,
131         doctor_id INT,
132         patient_name VARCHAR(50),
133         fee INT,
134         FOREIGN KEY (doctor_id)
135             REFERENCES doctors(doctor_id)
136             ON DELETE CASCADE
137     );
```

```
4  17:45:09  SHOW TABLES                                                    1 row(s) returned                0.000 sec / 0.000 sec
5  17:45:48  CREATE TABLE doctors (  doctor_id INT PRIMARY KEY,  dname VARCHAR(5...  0 row(s) affected      0.015 sec
6  17:45:58  CREATE TABLE appointments (  appt_id INT PRIMARY KEY,  doctor_id INT,  ...  0 row(s) affected  0.093 sec
```

Q2. Insert 3 doctors: (1, Dr. Sharma, Heart), (2, Dr. Rai, Bone), (3, Dr. KC, Eye).

```
140 •   INSERT INTO doctors
141     VALUES
142     (4, 'Dr. Sharma', 'Heart'),
143     (5, 'Dr. Rai', 'Bone'),
144     (6, 'Dr. KC', 'Eye');
145
```

```
17  17:52:42  INSERT INTO doctors VALUES (4, 'Dr. Sharma', 'Heart'), (5, 'Dr. Rai', 'Bone'), (6, 'D...  3 row(s) affected Records: 3  Duplicates: 0  Warnings: 0  0.000 sec
```

Q3. Try to insert a doctor with doctor ID = 2 again. Write down the error.

```
147 •   INSERT INTO doctors
148     VALUES (2, 'Dr. Thapa', 'Skin');
```

```
18  17:54:19  INSERT INTO doctors VALUES (2, 'Dr. Thapa', 'Skin')  Error Code: 1062. Duplicate entry '2' for key 'doctors.PRIMARY'  0.000 sec
```

Q4. Try to insert a doctor with NULL ID, and one with NULL name. Which constraint blocks each?

```
150 •   INSERT INTO doctors
151     VALUES (NULL, 'Dr. Hari', 'ENT');
```

```
19  17:56:05  INSERT INTO doctors VALUES (NULL, 'Dr. Hari', 'ENT')   Error Code: 1048. Column 'doctor_id' cannot be null   0.000 sec
```

Q5. For each real system, decide a good Primary Key and say why name is not enough: patients in this hospital, products in Daraz, citizens of Nepal.

System	Suitable primary key
Hospital patients	patient_id
Daraz products	Product_id
Nepal citizens	Citizens_no

Q6. Insert 4 appointments for existing doctors, two of them for doctor 1. Why is the repeated doctor ID allowed?

```
156     (101, 1, 'Ram', 1000),
157     (102, 1, 'Sita', 1500),
158     (103, 2, 'Hari', 1200),
159     (104, 3, 'Gita', 2000);
160
161 •   SELECT * FROM appointments;
```

Result Grid | Filter Rows: | Edit: | Export/Impo

appt_id	doctor_id	patient_name	fee
101	1	Ram	1000
102	1	Sita	1500
103	2	Hari	1200
104	3	Gita	2000
NULL	NULL	NULL	NULL

```
22  18:08:27  SELECT * FROM appointments LIMIT 0, 1000   4 row(s) returned
```

Q7. Try to insert an appointment with doctor ID = 99. What happens and why is this good?

```
163 •   INSERT INTO appointments
164     VALUES (105, 99, 'Nabin', 1000);
```

```
23  18:10:37  INSERT INTO appointments VALUES (105, 99, 'Nabin', 1000)   Error Code: 1452. Cannot add or update a child row: a foreign key constraint fails (h...  0.01
```

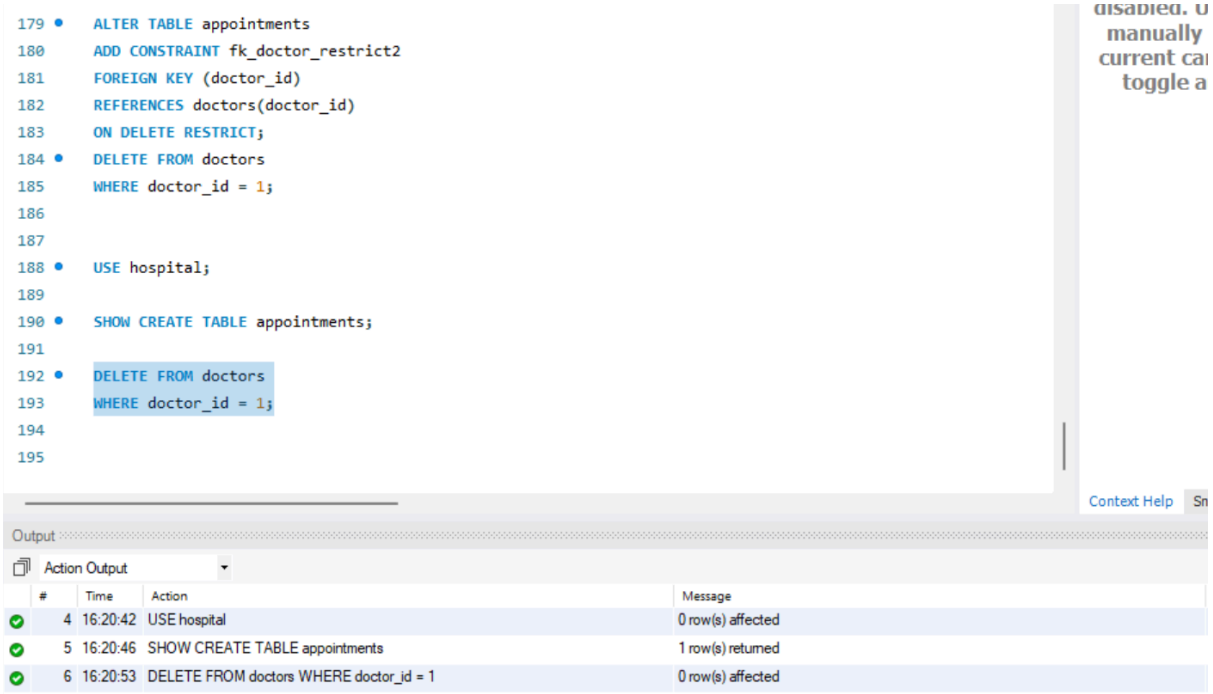
Q8. Which table is parent and which is child? Where does the Foreign Key live?
Parent table = doctors
Child table = appointments

Foreign Key = appointments.doctor_id
References = doctors.doctor_id

Q9. Delete doctor 1. What happens to doctor 1's appointments, and which Foreign Key option caused that?



Q10. Recreate the Foreign Key with RESTRICT instead. Try deleting a doctor who has appointments. What happens?



Q11. Real life: Daraz has customers and orders. Draw the parent, child, and Foreign Key column.

customers (Parent)
——----------------------------------
customer_id PRIMARY KEY
name
phone
      |

orders (Child)
----
order_id   PRIMARY KEY
customer_id FOREIGN KEY
product
Amount

Q12. Why can a Foreign Key repeat and be NULL while a Primary Key cannot? Answer in one sentence.

A Foreign Key can repeat and can be NULL if allowed by the table definition, while a Primary Key must be unique and cannot be NULL.